Single-Stage Uncertainty-Aware Jersey Number Recognition in Soccer

Łukasz Grad[1,2]

[1]University of Warsaw, Poland, l.grad@mimuw.edu.pl

[2]ReSpo.Vision, Poland, lukasz@respo.vision

Abstract

We present a single-stage uncertainty-aware approach for jersey number recognition in soccer. Our method employs digit-compositional classifiers that leverage the structural relationships between digits, coupled with a Dirichlet-based uncertainty modeling framework and a tracklet aggregation that combines frame-level predictions using confidence-based filtering. Unlike previous approaches that focus primarily on detecting visible jerseys, we reframe the problem through the lens of uncertainty quantification. This perspective enables more nuanced predictions, particularly in challenging cases of partial visibility or occlusion. Our unified architecture directly processes player crops without requiring explicit jersey detection as in traditional multi-stage pipelines. Through extensive experiments on the SoccerNet and Copa America (CA, ours) datasets, we demonstrate that digit-compositional approaches consistently outperform independent classifiers, while Dirichlet-based uncertainty modeling further improves performance by providing better calibrated confidence estimates across visibility conditions. We achieve high performance on the SoccerNet Challenge benchmark with 85.62% tracklet-level accuracy.

1. Introduction

Jersey number recognition is a crucial task in sports video analysis, enabling automatic player identification for tracking and game analytics. In team sports like soccer, players of the same team appear nearly identical except for their jersey numbers, making these numbers a key discriminative feature for player identification and re-identification [9].

Previous approaches have treated jersey number recognition as a two-stage pipeline: first detect or localize the jersey number region (either by detecting the player and then the number, or by isolating digits), then recognize the number via optical character recognition (OCR) or classification [8, 14]. While effective in controlled settings, such multi-stage systems face significant limitations. A critical drawback is the difficulty of quantifying confidence across the entire pipeline, as errors in early stages can propagate and impact subsequent recognition, a challenge highlighted by issues discussed in [2, 14]. The sequential nature of these systems makes it challenging to determine the reliability of the overall prediction, especially in ambiguous cases with partial visibility. Additionally, such pipelines require careful tuning of multiple components, with each stage needing separate training and optimization.

Recent research has begun to explore more integrated solutions. For instance, single-stage models [1, 23] aim to directly infer the jersey number from player images/tracklets in one end-to-end network, simplifying the pipeline. However, like many deep learning approaches, these methods can be susceptible to overfitting, especially when dealing with variations in labeling strategies (e.g., frame-level vs. tracklet-level). When processing entire player crops, models may learn to recognize specific players rather than their jersey numbers. Moreover, maintaining accuracy under real-world conditions (blur, low resolution, occlusion) remains challenging [13], often requiring region proposals.

In this work, we address these challenges by proposing a novel jersey number recognition model that leverages the compositional structure of digits while incorporating robust uncertainty estimation. Our approach utilizes a deep vision backbone [7] to extract powerful features from player detections. A key innovation lies in our jersey number head designs, which explicitly model the compositional nature of two-digit numbers (and handle single digits). This digit-compositional architecture encourages the model to focus on number structure rather than player identity, potentially mitigating overfitting issues seen in previous single-stage methods, and promoting generalization to unseen numbers. Furthermore, to handle ambiguous cases common in real-world footage (e.g., motion blur, occlusion), we integrate an evidence-based uncertainty modeling technique. Following the evidential deep learning framework [21], the network outputs parameters for a Dirichlet distribution over possible jersey numbers. This allows the model not only to predict a number but also to quantify its confidence via this distribution, enabling more reliable downstream decision-making (e.g., abstaining when uncertainty is high).

Integrating these components, our proposed single-stage, end-to-end trainable system takes player detections as input and directly outputs the jersey number and associated uncertainty, eliminating intermediate steps. In summary, the key contributions of this paper include:

- **Digit-Compositional Architecture:** A set of jersey number heads that leverages the compositional nature of digits, enabling understanding of structural relationships between numbers.
- **Uncertainty-Aware Predictions:** Integration of a Dirichlet-based evidential framework to provide uncertainty quantification alongside predictions, improving robustness in ambiguous cases.
- **Single-Stage Integration:** A unified, end-to-end trainable model that directly recognizes jersey numbers from player images, incorporating the digit-compositional heads and uncertainty estimation without requiring a separate text detection stage.

Following this introduction, Section 2 reviews relevant literature, including traditional jersey number recognition pipelines and recent advances in uncertainty estimation for vision. Sections 3 and 4 detail our datasets, model architecture with jersey number head designs, uncertainty modeling framework, and experimental results on benchmark datasets. Section 5 concludes the paper and discusses future work.

2. Related Work

2.1. Jersey Number Recognition in Sports

Early work on jersey number recognition often divided the problem into multiple sub-tasks. Classic pipelines detected the player or jersey region first, then applied OCR or a classifier to recognize the digits. For example, Gerke et al. used a conventional player detector and then a CNN to classify the jersey number from an image patch containing the player's upper body [8]. This two-stage strategy laid the groundwork, but suffered when jersey patches were low-resolution or partly occluded [8].

Subsequent research improved each stage by refining detection and localization of numbers. Liu and Bhanu introduced a pose-guided R-CNN, using human pose estimation to guide a region proposal network towards the back-number area before recognition [15]. Others applied instance segmentation or Mask R-CNN on players to better isolate jersey digits [16], effectively treating number recognition as a specialized case of scene text detection on player uniforms.

An alternative two-stage paradigm treats jersey numbers explicitly as text. Nady and Hemayed [19], and Chen and Poullis [2], explored using scene text detection algorithms to find jersey numbers, then reading them with a text recognizer. These text-detection methods tap into advances in generic scene text recognition and have shown promising results, especially with variable fonts or in sports like hockey where fonts can be highly stylized [2]. However, they require extensive annotation of digit bounding boxes and can misidentify unrelated text as numbers. Moreover, the two-stage nature persists: both detection and recognition networks must perform well for the system to succeed.

Single-stage methods aim to simplify this pipeline by internalizing the detection step. Li et al. pioneered an end-to-end trainable model using a Spatial Transformer Network to zoom in on the jersey number within the player image, followed by a CNN classifier [14]. Vats et al. proposed multi-task learning on player images, training a network to predict both the entire jersey number and each digit as separate classes [23], improving generalization despite focusing on hockey with a private dataset.

In the SoccerNet 2023 Jersey Number Challenge – the first large-scale public benchmark for this task – Balaji et al. developed a model that selects keyframes where the number is visible and applies a spatio-temporal network to predict the jersey number from the tracklet [1]. While this approach aggregates temporal information within a unified model, it still requires a keyframe selection module.

Overall, two-stage methods have dominated in challenging imagery scenarios but require greater complexity and inference time. Our work differs from prior single-stage approaches by explicitly accounting for prediction uncertainty using a principled evidential framework. By producing a distribution over possible numbers, our model expresses uncertainty in a principled way (e.g., between similar-looking numbers like "8" and "3" in blurry frames), leveraging recent advances in evidential deep learning.

2.2. Uncertainty Modeling with Dirichlet Distribution in Vision

In computer vision applications, knowing when not to trust a prediction can be as important as the prediction itself. Traditional neural networks often yield over-confident predictions even on unfamiliar inputs [11]. While Bayesian neural networks or ensemble methods can estimate confidence, they tend to be computationally expensive.

Evidential deep learning produces uncertainty estimates in a single forward pass by modeling the output as a distribution over class probabilities [21]. Sensoy et al. [21] introduced a framework where a network predicts "evidence" for each class, parametrizing a Dirichlet distribution. The total evidence corresponds to confidence: strong evidence for one class creates a peaked Dirichlet (low uncertainty), while low evidence produces a flat distribution (high uncertainty). This approach allows networks to learn uncertainty from data by penalizing over-confidence on wrong predictions.

Dirichlet-based uncertainty modeling has been applied

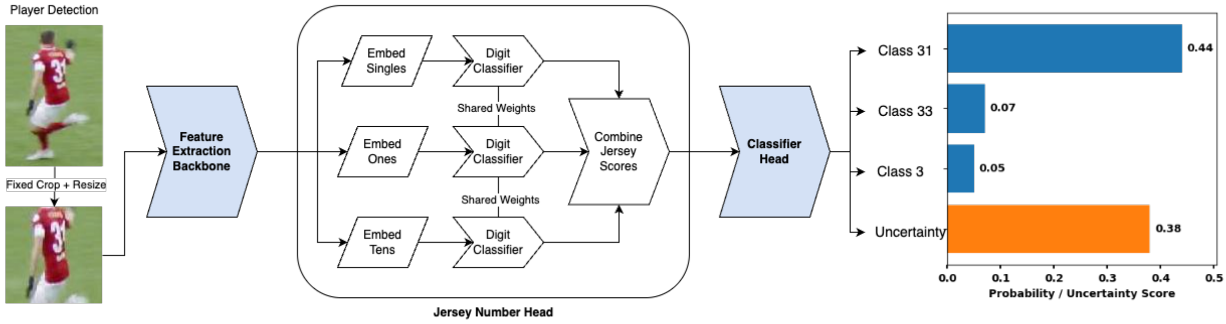

Figure 1. Overview of the proposed jersey recognition architecture. A pre-trained backbone (f_{backbone}) extracts features from player crops. The Jersey Number Head (f_{jersey}), specifically the Tied Digit-aware (TDA) variant shown here, processes these features using position embeddings and a shared digit classifier to produce jersey scores. The final Classification Head (f_{class}) transforms these scores into probabilities and an uncertainty estimate using an uncertainty-aware mechanism (e.g., Dirichlet). We present an actual output from ViT-B/8 trained on 200M dataset.

Table 1. Quantitative summary of datasets employed in our jersey number recognition experiments.

Dataset	Tracklets	Images
SoccerNet Train	1,427	733,000
SoccerNet Test	1,211	564,500
SoccerNet Challenge	1,426	748,600
SoccerNet ReID	–	350,986
200M	594,977	1,071,564
Copa America (CA)	33,730	89,728

to various vision tasks beyond classification. Malinin and Gales [18] proposed Prior Networks with explicit Dirichlet priors to represent different uncertainty levels, while Kendall and Gal [12] underscored the importance of modeling both aleatoric and epistemic uncertainty. Evidential approaches capture this blend without requiring sampling during inference.

For jersey number recognition, uncertainty estimation is particularly valuable. When jersey numbers are not visible (e.g. player facing towards camera, partial occlusion), deterministic models might still output a number due to learned biases, while uncertainty-aware models can express low confidence. Leveraging a Dirichlet-based model, our system flags low-confidence outputs, preventing misidentifications in tracking by deferring decisions on ambiguous frames.

3. Methodology

3.1. Datasets

Table 1 presents a quantitative overview of the datasets used in this study. We utilize both publicly available SoccerNet datasets and our proprietary datasets. The following subsections detail the characteristics and annotation methodologies for each.

3.1.1. SoccerNet Datasets

The SoccerNet jersey number recognition datasets serve as the primary public benchmark for this task [3–5, 10].

Official SoccerNet Splits We use three official splits:

- **SoccerNet Train**: The standard training set, used alongside the pseudolabeled SoccerNet ReID data for training models on public data only.
- **SoccerNet Test**: Used for internal testing and validation.
- **SoccerNet Challenge**: Used for final evaluation and comparison with state-of-the-art methods.

An important feature of these SoccerNet splits is the provision of explicit tracklet-level jersey visibility annotations. Furthermore, tracklets with an average diagonal smaller than 40 pixels were excluded from training or assigned label 1 (during inference), mitigating noise from known errors where small bounding boxes corresponded to ball detections rather than players.

SoccerNet ReID Dataset This dataset originates from the SoccerNet ReID challenge and contains diverse player crops without tracking information. We applied pseudolabeling to this dataset using our best model trained on proprietary data. An uncertainty-aware approach was used: images with low predicted uncertainty were labeled with the predicted number, high uncertainty images were labeled as 'absent', and intermediate uncertainty images were discarded (details in Section 8.1). This process resulted in a refined dataset of 350,986 images, enabling the training of competitive models using only public data.

Figure 2. Qualitative results on the Copa America (CA) dataset. Each panel shows a player crop with ground truth (GT) and predicted (Pred) detection-level jersey number. The border color indicates prediction uncertainty, mapped by the color bar. Note the correlation between higher uncertainty (orange/red borders, bottom row) and challenging images, compared to lower uncertainty (blue borders, top row) on clearer, often correctly predicted examples. Results are from our ViT-B/8 trained on 200M dataset. See Figures 5 and 6 in Supplementary Material for more examples.

3.1.2. Proprietary Datasets

We utilize two large-scale proprietary datasets, 200M and Copa America (CA), which feature distinct teams and players with no overlap between them.

200M Dataset This is our primary training dataset, extracted from over 200 unique soccer matches across various leagues and competition levels (professional, youth, women). It features weak labeling at the tracklet level, where annotations are assigned per tracklet. This was performed by expert annotators following a rigorous protocol involving continuous player tracking review and a dedicated quality assurance step achieving 99.5% accuracy. Frames with significant occlusions were systematically filtered during this process to ensure data quality.

Copa America (CA) Dataset This dataset serves as a secondary test set, containing player crops from 12 matches of the Copa America 2024 tournament (see Figure 2 for examples). It is used to evaluate model generalization to unseen jersey designs and conditions. It employs the same weak tracklet-level annotation methodology as the 200M dataset.

3.1.3. Legibility Pseudolabeling for Proprietary Data

The 200M and CA datasets lack frame-level visibility annotations. To address this, we employed a proprietary jersey number detector to generate pseudolabels indicating whether a jersey number is likely legible in each frame. A relatively low confidence threshold was used in the detector to determine legibility, resulting in approximately 30% of frames being labeled as 'legible'. While this heuristic may include some false positives, it effectively captures many challenging frames with partially visible numbers, enriching the training data. Although the tool used is proprietary, the concept of using an auxiliary model or heuristic to pseudolabel visibility is generally applicable.

3.1.4. Data Preprocessing

Our method operates directly on player bounding box crops. The core preprocessing, applied during both training and inference, involves cropping the torso region (removing the top 1/6 and bottom 1/3 of the bounding box) and resizing the result to a fixed resolution (e.g., 224x224 pixels). This location-focused preprocessing is independent of the recognition model itself.

During training only, we apply standard data augmentations, including random scaling, color jitter, random rotations, and variations in brightness and blur. Specific augmentation parameters (e.g., rotation degrees, scaling ranges) are detailed in the Supplementary Material (Section 7.3).

3.1.5. Dataset Characteristics Comparison

The datasets present distinct characteristics impacting model training and evaluation. SoccerNet generally features longer tracklets and higher average resolution player crops compared to our proprietary datasets (200M and CA). SoccerNet also provides explicit tracklet-level visibility labels, unlike the proprietary sets which rely on weak tracklet-level number annotations and pseudolabeled frame legibility.

The shorter tracklets and smaller, lower-resolution crops typical in the 200M and CA datasets increase the recognition challenge but arguably better reflect variability in real-world broadcast footage where players appear at diverse distances and angles. These differences highlight the importance of evaluating models across multiple datasets with varying properties. A more detailed comparative analysis, including visual examples, is provided in the Supplementary Material (Section 8).

3.2. Jersey Recognition Architecture

3.2.1. Overview of proposed approach

The system consists of three primary components:

1. **Feature Extraction Backbone**: A pre-trained model that processes raw player images $\mathbf{I}$ and extracts visual features $\mathbf{x} = f_{\text{backbone}}(\mathbf{I})$. The architecture supports any feature extractor that produces image embeddings.
2. **Jersey Number Head**: A component that maps visual features to jersey scores $\mathbf{s} = f_{\textbf{jersey}}(\mathbf{x})$, with implementations ranging from independent class treatment to digit-compositional models. The head returns a 100-dimensional vector for Dirichlet-based models or a 101-dimensional vector (including absent class) for softmax-based models.
3. **Classification Head**: The final component that transforms the jersey scores into probabilities and uncertainty estimates $p, u = f_{\text{class}}(\mathbf{s})$, responsible for the prediction and confidence quantification.

The primary contribution in our approach is the integration of these components into a single trainable model that directly processes player crop images without requiring explicit jersey detection as a preprocessing step, see Figure 1 for an overview. In our implementation, we use the CLS ViT [7] token output as the global image embedding, producing a feature vector of dimension D that serves as input to the classification head.

3.2.2. Jersey Number Head

We investigate three approaches to jersey number representation that differ in parameter efficiency and digit compositionality:

Independent Head (IND). This approach treats each jersey number as a separate class, mapping image features directly to 101 scores (numbers 0-99 plus an absent class) using a single linear layer:

$$\mathbf{s} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

where $\mathbf{W} \in \mathbb{R}^{101\times D}$ is the weight matrix and $\mathbf{b} \in \mathbb{R}^{101}$ is the bias vector. With D dimensional embeddings, this requires $101D + 101$ parameters.

Digit-aware Head (DA). This approach decomposes the problem into digit-level classification, producing scores using separate linear layers for single digits ($\mathbf{s}_{\text{single}}$), tens place ($\mathbf{s}_{\text{tens}}$), ones place ($\mathbf{s}_{\text{ones}}$), and an absent class ($s_{\text{absent}}$):

$$\mathbf{s}_p = \mathbf{W}_p\mathbf{x} + \mathbf{b}_p$$

Note that $\mathbf{s}_{\text{single}}, \mathbf{s}_{\text{tens}}, \mathbf{s}_{\text{ones}} \in \mathbb{R}^{10}$ while $s_{\text{absent}} \in \mathbb{R}$. The final scores for the 101 output classes (0-99 plus absent) are constructed as follows: scores for single-digit numbers $j \in [0, 9]$ are taken directly from the corresponding j-th element of $\mathbf{s}_{\text{single}}$. Scores for two-digit numbers $10 \cdot t + o$ (where $t, o \in [0, 9]$ and $t \neq 0$) are constructed by adding the corresponding tens and ones digit scores: $s_{10\cdot t+o} = s_{\text{tens},t} + s_{\text{ones},o}$. The score for the absent class is s_{absent}. This approach requires $31D + 31$ parameters, reducing the parameter count compared to IND while enabling knowledge sharing within digit positions.

Tied Digit-aware Head (TDA). Our proposed approach shares weights across digit positions, using learnable position embeddings to distinguish between single, tens, and ones places:

$$\mathbf{x}_p = \begin{cases} \mathbf{x} + \mathbf{e}_p & \text{for additive embeddings} \\ \mathbf{x} \odot \mathbf{e}_p & \text{for multiplicative embeddings} \end{cases}$$

where $\mathbf{x}_p$ represents features transformed for position $p \in \{\text{single}, \text{tens}, \text{ones}\}$, $\mathbf{e}_p \in \mathbb{R}^D$ is the learnable position embedding for position p, and $\odot$ represents element-wise multiplication. All digit scores are computed using a shared weight matrix:

$$\mathbf{s}_p = \mathbf{W}_{\text{digit}}\mathbf{x}_p + \mathbf{b}_p$$

where $\mathbf{s}_p$ represents scores for position p. This approach offers two key variants: additive vs. multiplicative embeddings and per-digit vs. per-position biases. With either variant, the parameter count is $14D + 31$ (with per-digit bias) or $14D + 4$ (with per-position bias), representing an 86% reduction compared to the Independent Head.

Parameter Efficiency. The parameter counts for embedding dimension D are summarized in Table 2. Both the DA and TDA heads provide significant advantages: (1) knowledge sharing across positions, (2) compositional understanding of numbers, and (3) improved generalization to rare or unseen number combinations. Additionally, their reduced parameter space limits overfitting to training data, enabling more robust recognition.

Table 2. Parameter comparison of jersey number head variants

Head Type	Configuration	Parameter Count
IND	Standard	$101D + 101$
DA	Standard	$31D + 31$
TDA	Per-Digit Bias	$14D + 31$
TDA	Per-Position Bias	$14D + 4$

3.2.3. Uncertainty Modeling

We investigate two approaches for modeling uncertainty in jersey number recognition: softmax-based and Dirichlet-based methods.

Softmax-Based Approach. This method implements uncertainty through an additional "absent" class (index 100). The jersey scores $\mathbf{s}$ from the head are treated as logits and converted to a probability distribution $\mathbf{p} = \text{Softmax}(\mathbf{s})$, where p_j is the probability for class j. In this formulation, the probability assigned to the absent class serves as the uncertainty measure. The model is trained with cross-entropy loss $\mathcal{L}_{\text{CE}} = -\sum_{j=0}^{100} y_j \log(p_j)$, where $y_j = (1 - \epsilon)$ for the true class and $y_j = \epsilon/(N-1)$ for other classes, with $\epsilon = 10^{-2}$ and $N = 101$. This label smoothing accounts for potential label noise in the datasets. Samples without visible jersey numbers are mapped to the absent class. This approach enforces mutual exclusivity between predictions of specific numbers and the absent class.

Dirichlet-Based Approach. We propose an alternative approach based on Dirichlet distributions [21], interpreting network outputs as evidence for each class. For a feature vector $\mathbf{x}$, the model produces jersey scores $\mathbf{s}(\mathbf{x}|\theta) \in \mathbb{R}^{100}$ from the head, which are then transformed into Dirichlet concentration parameters (evidence):

$$\alpha_j = \exp(s_j(\mathbf{x}|\theta)) + 1$$

where α_j represents evidence for class j. The total evidence $S = \sum_{k=0}^{99} \alpha_k$, expected probability $p_j = \alpha_j / S$, and uncertainty score uncertainty $= 100/S$. For training, we employ the Type-II Maximum Likelihood loss as derived in [21]:

$$\mathcal{L}_{\text{ML}} = \sum_{j=0}^{99} y_j \cdot (\log(S) - \log(\alpha_j))$$

For samples without visible jersey numbers, we apply KL regularization that encourages a uniform distribution across classes:

$$\mathcal{L}_{\text{KL}} = \text{KL}(\text{Dir}(\alpha) \| \text{Dir}(\mathbf{1}))$$

where $\text{Dir}(\mathbf{1})$ represents a uniform Dirichlet distribution. The total loss combines both terms:

$$\mathcal{L}_{\text{Dir}} = \mathcal{L}_{\text{ML}} + \lambda_{\text{KL}} \cdot \mathcal{L}_{\text{KL}}$$

where λ_{KL} controls the strength of regularization. The value of λ_{KL} follows a specific schedule during training, detailed in Section 7 of the Supplementary Material.

The two approaches differ in how they represent uncertainty (explicit absent class via softmax probabilities vs. evidence concentration via Dirichlet parameters), handle loss computation, process invisible numbers, and make predictions.

3.2.4. Tracklet Aggregation

In sports video analysis, players are tracked across consecutive frames, forming tracklets. While our model makes predictions for individual frames, the final goal is to assign a single jersey number to each player tracklet. We present an uncertainty-aware approach for aggregating per-frame predictions into tracklet-level predictions.

Our tracklet aggregation [1, 13] method combines predictions across frames by considering both uncertainty and class probabilities. The key parameters are:

1. **Invisible threshold $\tau_{\text{invisible}}$**: If the minimum uncertainty across all frames in a tracklet exceeds this threshold, the entire tracklet is classified as having no visible jersey number.
2. **Uncertainty threshold $\tau_{\text{uncertainty}}$**: Only frames with uncertainty below this threshold contribute to the tracklet-level prediction.

We combine predictions across the remaining frames using a weighted average of probabilities, incorporating a Bayesian-inspired prior with astrength parameter α. The tracklet-level probability distribution is computed as:

$$p_{\text{tracklet}} = \frac{\sum_{i \in F} p_i + \alpha \cdot p_{\text{prior}}}{|F| + \alpha}$$

where F is the set of frames with uncertainty below $\tau_{\text{uncertainty}}$, p_i is the probability distribution for frame i, p_{prior} is a uniform prior distribution, and $|F|$ is the cardinality of set F. The final jersey number prediction is the class with maximum probability in p_{tracklet}. The dataset-specific threshold values used and an analysis of their sensitivity are detailed in the Supplementary Material (Section 9.2).

4. Results

4.1. Experimental Setup

To evaluate our approach, we conducted experiments across multiple datasets, classifier architectures, and uncertainty modeling configurations. We utilized three primary datasets: our proprietary 200M dataset, SoccerNet Test, and CA. For public benchmarking, we additionally evaluated on the SoccerNet Challenge set [4].

Our ablation studies are based on a Vision Transformer Small (ViT-S/16) [7] backbone pre-trained on ImageNet-21k [6, 20], with input size of 224×224 pixels and feature dimension of 384. For final benchmarks, we also utilized Vision Transformer Base (ViT-B/8) models. We experimented with all jersey head variants (IND, DA, TDA) and both uncertainty modeling approaches (softmax-based and Dirichlet-based). For tracklet-level predictions, we employed the uncertainty-based aggregation approach described in Section 3.2.4.

We evaluated models using tracklet-level accuracy, reporting both overall accuracy and separate metrics for visible and invisible jerseys where applicable. To ensure statistical significance, all experiments were conducted with three independent training runs using different random seeds. Complete training configuration details, including optimizer settings, learning rate schedules, and regularization techniques, are provided in the supplementary material Section 7.

4.2. Ablation Studies

4.2.1. Jersey Number Head Comparison

To assess the effectiveness of our different jersey number head approaches, we compared all variants on both SoccerNet Test and CA datasets.

The detailed comparison of jersey number heads on the SoccerNet Test dataset can be found in Table 4, which presents results for both softmax and Dirichlet-based approaches. The Digit-Aware (DA) head achieves the highest overall accuracy, outperforming the Independent (IND) head by a significant margin. Among the Tied Digit-Aware variants, the multiplicative embedding (TDA-M) consistently outperforms the additive embedding (TDA-A).

Results on the more challenging CA dataset, presented in Table 5, exhibit a similar pattern. When examining the softmax-based approach, TDA-M achieves the highest accuracy, followed closely by TDA-MB (best for Dirichlet) and DA. The IND head consistently performs worst, confirming the advantage of digit-compositional approaches across different data distributions.

All models perform similarly on invisible jersey detection in SoccerNet, suggesting this capability depends more on uncertainty modeling than head architecture. The digit-compositional approaches not only improve performance but also substantially reduce head parameters. The TDA-M head provides the optimal balance between efficiency and accuracy, validating our hypothesis that understanding the compositional nature of numbers is critical for effective jersey recognition.

4.2.2. Hold-out Number Experiment

To evaluate the compositional understanding and generalization capabilities of our jersey number heads, we designed a challenging hold-out experiment on the SoccerNet Test dataset. We systematically removed three specific jersey numbers (one single-digit and two two-digit numbers) from the training set. We evaluated the models' ability to recognize these unseen numbers using balanced accuracy to mitigate potential class imbalance within the held-out set. During prediction, we restricted the model to predict only the held-out numbers, masking all other classes.

Table 3 presents the results. All digit-compositional approaches (DA, TDA variants) significantly outperform the Independent (IND) head, demonstrating a substantial capability to recognize numbers never encountered during training. Comparing uncertainty modeling approaches, the Dirichlet-based models generally achieve higher masked balanced accuracy than their softmax counterparts, with TDA-M showing the best performance overall. The IND head struggles considerably with both softmax and Dirichlet, confirming its inability to generalize compositionally.

Table 3. Hold-out number experiment on SoccerNet Test using Balanced Accuracy. Head type abbreviations as in Table 4.

Head Type	Softmax (%)	Dirichlet (%)
TDA-M	**77.93 $\pm$ 12.73**	**85.86 $\pm$ 13.88**
TDA-MB	74.38 $\pm$ 13.52	84.11 $\pm$ 13.59
TDA-A	69.21 $\pm$ 18.85	68.00 $\pm$ 23.22
DA	60.84 $\pm$ 19.71	65.68 $\pm$ 15.99
IND	35.59 $\pm$ 15.78	22.49 $\pm$ 15.15

The consistently high standard deviations across most methods, particularly TDA-A and DA, indicate that performance varies significantly depending on which specific numbers are held out. This suggests that certain number combinations pose greater challenges for zero-shot generalization than others. Nevertheless, the strong performance of the digit-aware models, especially TDA-M, provides compelling evidence that they develop a fundamental understanding of number composition.

4.2.3. Uncertainty Modeling Impact

To evaluate the effectiveness of our Dirichlet-based uncertainty modeling approach compared to the standard softmax-based method, we conducted experiments with all five head variants. This comparison allows us to assess the impact of uncertainty modeling on overall performance and generalization capability.

Table 4 presents results on the SoccerNet Test dataset. The Dirichlet-based approach significantly outperforms the softmax-based method across four of the five head types. Notably, the improvement in invisible jersey detection is particularly substantial across all head types.

Results on the more challenging CA dataset, shown in Table 5, reveal a consistent pattern of improvement with Dirichlet-based uncertainty modeling. All head types benefit from this approach. The consistent improvement indicates that the benefits of evidential uncertainty modeling are robust across different data distributions.

The Dirichlet-based uncertainty modeling approach demonstrates key advantages over the traditional softmax method. Most notable is the substantial improvement in invisible jersey detection, suggesting that evidence-based uncertainty quantification excels at determining jersey visibility. The Independent head sees one of the largest accuracy boosts, indicating that uncertainty modeling can partially compensate for less compositional architectures.

4.2.4. SoccerNet Challenge Results

To establish the effectiveness of our approach in comparison to existing methods, we evaluated our best-performing models on the official SoccerNet jersey number recognition

Table 4. Uncertainty modeling impact on SoccerNet Test dataset. DA: Digit-Aware, TDA-M: Tied Digit-Aware with Multiplicative embedding, TDA-MB: TDA-M with Per-Digit Bias, TDA-A: TDA with Additive embedding, IND: Independent.

Head	Acc Total (%)		Acc Vis (%)		Acc Invis (%)	
Type	Softmax	Dirichlet (Δ)	Softmax	Dirichlet (Δ)	Softmax	Dirichlet (Δ)
DA	**80.15** $\pm$ **1.16**	**83.24** $\pm$ **0.94** (+3.09)	**83.72** $\pm$ **0.85**	**85.83** $\pm$ **1.17** (+2.11)	71.55 $\pm$ 2.46	77.00 $\pm$ 1.07 (+5.45)
TDA-M	78.70 $\pm$ 0.22	82.91 $\pm$ 0.25 (+4.21)	81.85 $\pm$ 0.82	85.71 $\pm$ 0.70 (+3.86)	71.08 $\pm$ 2.29	76.15 $\pm$ 0.86 (+5.07)
TDA-MB	77.81 $\pm$ 0.59	82.11 $\pm$ 1.92 (+4.30)	80.49 $\pm$ 1.52	84.11 $\pm$ 3.65 (+3.62)	71.36 $\pm$ 2.40	77.28 $\pm$ 2.28 (+5.92)
TDA-A	75.81 $\pm$ 0.73	73.03 $\pm$ 0.80 (-2.78)	77.18 $\pm$ 1.66	71.11 $\pm$ 0.53 (-6.07)	**72.49** $\pm$ **1.55**	**77.65** $\pm$ **1.55** (+5.16)
IND	73.47 $\pm$ 1.25	78.48 $\pm$ 0.55 (+5.01)	74.61 $\pm$ 0.97	79.05 $\pm$ 1.94 (+4.44)	70.70 $\pm$ 1.95	77.09 $\pm$ 3.26 (+6.39)

Table 5. Uncertainty modeling impact on CA dataset. Head type abbreviations as in Table 4.

Head Type	Softmax (%)	Dirichlet (Δ) (%)
DA	26.33 $\pm$ 0.19	27.32 $\pm$ 0.15 (+0.99)
TDA-M	**26.89** $\pm$ **0.27**	27.91 $\pm$ 0.20 (+1.02)
TDA-MB	26.60 $\pm$ 0.26	**28.07** $\pm$ **0.54** (+1.47)
TDA-A	25.57 $\pm$ 0.18	27.88 $\pm$ 0.17 (+2.31)
IND	24.90 $\pm$ 0.19	26.20 $\pm$ 0.82 (+1.30)

benchmark [4]. Table 6 presents our results alongside previous work, demonstrating significant advancements over the state-of-the-art.

Table 6. Comparison with prior work on the SoccerNet benchmarks.

Method	Test Acc	Challenge Acc
Gerke et al. [8]	32.57%	35.79%
Vats et al. [23]	46.73%	49.88%
Li et al. [14]	47.85%	50.60%
Vats et al. [24]	52.91%	58.45%
Balaji et al. [1]	68.53%	73.77%
Koshkina et al. [13]	87.45%	79.31%
Ours (ViT-S, SoccerNet)	82.74%	-
Ours (ViT-B, SoccerNet)	86.37%	83.52%
Ours (ViT-B, 200M)	85.46%	**85.62%**
Ours (ViT-B, finetuned)	**88.27%**	85.41%

Our best proprietary model, a ViT Base with patch size 8 trained on our 200M dataset, achieves 85.62% accuracy on the SoccerNet Challenge set, representing a substantial improvement over the previous highest published research results by Koshkina et al. [13] (79.31%). The finetuned variant of this model, which leverages one epoch of additional training on the SoccerNet Train set, achieves 88.27% on the SoccerNet Test set and 85.41% on the Challenge set, further validating our methodological approach.

Particularly noteworthy is our model trained exclusively on publicly available data (ViT-B, SoccerNet), which achieves 86.37% accuracy on the SoccerNet Test set and 83.52% on the Challenge set. This result demonstrates the effectiveness of our pseudolabeling approach for the SoccerNet ReID dataset described in Section 3.1.1. Even our smaller ViT-Small variant trained solely on public data achieves 82.74% accuracy on the SoccerNet Test set.

The consistent performance improvements across all model variants validate our key design choices: digit-compositional heads, Dirichlet-based uncertainty modeling, and uncertainty-aware tracklet aggregation. Moreover, the strong results from publicly-trained models indicate that our approach achieves high-performance without requiring access to large-scale proprietary datasets, thus enhancing accessibility and reproducibility of our research.

5. Conclusions

In this paper, we presented a single-stage, uncertainty-aware jersey number recognition approach employing digit-compositional classifiers and Dirichlet-based uncertainty modeling. Experiments confirmed the benefits of digit-aware heads, their generalization to unseen numbers, and improvements from the Dirichlet framework over softmax.

Our approach achieves very high performance on the SoccerNet Challenge benchmark, improving upon previous methods. This success validates our key design choices: decomposing numbers into constituent digits, modeling uncertainty through evidence accumulation, and employing uncertainty-aware tracklet aggregation.

Future work could explore extending our digit-compositional framework to other sports and different uniform styles. Investigating alternative uncertainty quantification methods beyond the evidence-based Dirichlet approach could also yield further improvements. Finally, incorporating temporal modeling for improved uncertainty estimation across tracklets represents a promising direction.

6. Acknowledgments

This work was co-financed by the Polish National Centre for Research and Development under Grant POIR.01.01.01-00-0718/21 as part of the project titled "Universal, automatic, artificial intelligence-based sports analytics system enabling acquisition & contextualization of 3D positional data from sports games and prediction of game events as well as assessment of tactics, skills and potential of teams & players".

References

[1] Bavesh Balaji, Jerrin Bright, Harish Prakash, Yuhao Chen, David A. Clausi, and John S. Zelek. Jersey number recognition using keyframe identification from low-resolution broadcast videos. In *Proc. 6th Intl. Workshop on Multimedia Content Analysis in Sports (ACM MMSports)*, 2023. 1, 2, 6, 8

[2] Qiao Chen and Charalambos Poullis. Tracking and identification of ice hockey players. In *Proc. Intl. Conf. on Computer Vision Systems (ICVS)*, 2023. 1, 2

[3] Anthony Cioppa, Adrien Deliège, Silvio Giancola, Bernard Ghanem, and Marc Van Droogenbroeck. Scaling up soccernet with multi-view spatial localization and re-identification. *Scientific Data*, 9:355, 2022. 3

[4] Anthony Cioppa, Christophe De Vleeschouwer, Vladimir Somers, and et al. SoccerNet 2023 challenges results. *Sports Engineering*, 27(2):1–18, 2024. 6, 8

[5] Adrien Deliège, Anthony Cioppa, Silvio Giancola, Meisam J. Seikavandi, Jacob V. Dueholm, Kamal Nasrollahi, Bernard Ghanem, Thomas B. Moeslund, and Marc Van Droogenbroeck. Soccernet-v2 : A dataset and benchmarks for holistic understanding of broadcast soccer videos. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2021. 3

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255. IEEE, 2009. 6

[7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, and et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021. 1, 5, 6

[8] Sebastian Gerke, Karsten Müller, and Ralf Schäfer. Soccer jersey number recognition using convolutional neural networks. In *Proc. ICCV Workshops*, pages 17–24, 2015. 1, 2, 8

[9] Sebastian Gerke, Antje Linnemann, and Karsten Müller. Soccer player recognition using spatial constellation features and jersey number recognition. *Computer Vision and Image Understanding*, 159:105–115, 2017. 1

[10] Silvio Giancola, Mohieddine Amine, Tarek Dghaily, and Bernard Ghanem. Soccernet: A scalable dataset for action spotting in soccer videos. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2018. 3

[11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proc. ICML*, pages 1321–1330, 2017. 2

[12] Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS) 30*, pages 5574–5584, 2017. 3

[13] Maria Koshkina and James H. Elder. A general framework for jersey number recognition in sports video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVSports)*, 2024. 1, 6, 8

[14] Gen Li, Shikun Xu, Xiang Liu, Lei Li, and Changhu Wang. Jersey number recognition with semi-supervised spatial transformer network. In *Proc. CVPR Workshops*, pages 1783–1790, 2018. 1, 2, 8

[15] Hengyue Liu and Bir Bhanu. Pose-guided r-cnn for jersey number recognition in sports. In *Proc. CVPR Workshops*, pages 2457–2466, 2019. 2

[16] Hengyue Liu and Bir Bhanu. JEDE: Universal jersey number detector for sports. *IEEE Trans. on Circuits and Systems for Video Technology*, 32(11):7894–7909, 2022. 2

[17] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. 1

[18] Andrey Malinin and Mark Gales. Predictive uncertainty estimation via prior networks. In *Advances in Neural Information Processing Systems (NeurIPS) 32*, pages 7047–7058, 2018. 3

[19] Ahmed Nady and Elsayed E. Hemayed. Player identification in different sports. In *Proc. Intl. Conf. on Computer Vision Theory and Applications (VISAPP)*, pages 653–660, 2021. 2

[20] Tal Ridnik, Emanuel Ben-Baruch, Asaf Noy, and Lihi Zelnik-Manor. Imagenet-21k pretraining for the masses. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 6

[21] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems (NeurIPS) 31*, pages 3179–3189, 2018. 1, 2, 6

[22] Leslie N. Smith. Cyclical learning rates for training neural networks. In *2017 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 464–472, 2017. 1

[23] Kanav Vats, Mehrnaz Fani, David A. Clausi, and John S. Zelek. Multi-task learning for jersey number recognition in ice hockey. In *Proc. 4th Intl. Workshop on Multimedia Content Analysis in Sports (MMSports)*, pages 11–15, 2021. 1, 2, 8

[24] Kanav Vats, William McNally, Pascale Walters, David A. Clausi, and John S. Zelek. Ice hockey player identification via transformers and weakly supervised learning. In *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVSports)*, pages 3451–3460, 2022. 8